

Module 1: Introduction To Explainable Ai (Xai)

Instructional Hours: 4

Module Overview

In this module, you will learn about the need for responsible and explainable Artificial Intelligence (AI) in the Development of AI systems. You will learn about the key components of responsible AI. You will also learn about the techniques used to implement explainability in AI. You will also learn about the interaction of Responsible and Explainable AI.

Learning Outcomes

At the end of this module, you will be able to:

1. Define Responsible and Explainable AI
2. Discuss the Elements of Responsible and Explainable of AI
3. Classify Explainable AI Techniques used AI Systems
4. Justify the need for Responsible and Explainable in AI Systems

Learning Activities/task List

1. Review Introductory notes on Responsible and Explainable AI
2. Participate in an Online Discussion on the Motivation for Responsible and Explainable AI
3. Review Notes on Explainable AI Approaches
4. Complete a Quiz on Explainable AI Approaches
5. Review Notes on Elements of Responsible AI
6. Participate in an Online Discussion on the Elements of Explainable AI
7. Read Reference Materials on the Nexus of XAI and RAI
8. Participate in an Online Discussion on the Nexus of XAI and RAI
9. Complete a Final Module Quiz

Introduction To Responsible And Explainable Ai (Rxai)

Artificial intelligence (AI) has been adopted in many spheres of human life. It is ubiquitously being employed in decision making in a wide range of sectors such as health, law, education, finance and even in the military. In health, it is used for diagnosis and to recommend treatments. In law, it is used to assess suspects for recidivism and also identify suspects. In education, it is used in automatic grading and assessing candidates for university placement. In finance, it is used to assess a client for credit worthiness. In the military, it is used to select military targets to be attacked during war. In these applications, AI models make decision that can have a profound effect on human lives. It is therefore becoming increasingly necessary to comprehend how AI systems go about making these important decisions in order to determine if these decisions are justified (Arrieta et al., 2020).

According to Baker S. & Xiang S. (2023), just like any other tool, "AI can AI can also cause great harm if developed and used improperly or inappropriately". This harmfulness can be either intentional (malicious) or by accident (Baker S. & Xiang S. 2023). Examples of harm that can occur include bias in rejection of loan applications and job placement, flagging of a transaction as fraudulent while it isn't, incorrect diagnosis of diseases and in appropriate selection of a military target.

Responsible AI (RAI) emerged out the need to develop AI such that it is beneficial to humans and society in general (Baker S. & Xiang S. 2023). It involves applying ethics in the development and use of AI systems for the greater good of humanity. RAI has many components but some of the main ones that have been identified include fairness (or lack of bias), robustness (reliability), transparency, accountability, privacy and safety.

Explainability is related to transparency and is one component of RAI. Explainable AI (XAI) is "focused on providing humans with insight into the reasoning of AI models during their decision making processes" (Baker S. & Xiang S. 2023). Explanation is the ability of a system to generate accompanying evidence or reasons for outputs and/or processes (Phillips et al., 2021). Explainability can be considered as one of the components of responsible AI as it ensures transparency of the system. According to Baker S. & Xiang S. (2023), Explainability is a key pillar because it is useful in providing evidence of the attainment of all other components. This is illustrated in figure 1.1 below.

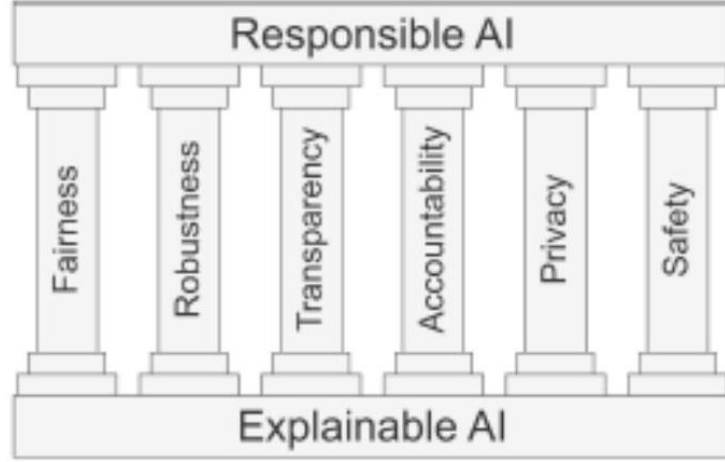


Figure 1.1: Relationship of RAI and XAI (Baker S. & Xiang S. 2023).

According to Baker S. & Xiang S. (2023), explainability is necessary because of the increasing complexity of AI models. Early models such as linear regression, decision trees and k -nearest neighbours are explainable because the mechanism by which they connect their inputs to the outputs is easy to understand and can be replicated by humans. The performance of these models however is not as good as more recent techniques like ensemble methods and deep learning based models. These recent methods have much better performance. However, the mechanism they use to transform their inputs to outputs is not inherently understandable to humans. These models are therefore sometimes referred to as black-box models. In order to increase trust in such models, it is necessary that their decision making criteria is revealed to humans. This has been described as contributing to the development of trustworthy AI Systems. If users perceive AI systems as trustworthy, then they are likely to have less resistance in adopting them. Black-box AI systems on other hand are likely to face higher adoption resistance.

In implementing XAI, it is always desirable to consider the audience for whom the explanation is generated. The audience usually consists of developers and end users. For the developers, explanations should be geared towards helping the debug and improve the models. For the end users, the goal is to consider if the system is considering the right factors or attributes in decision making. The explanations generation should be meaningful and accurate (Phillips et al., 2021).

The need for XAI may also arise out of legal and regulatory requirements for reasons to be given for decisions, including those made by automated systems. Examples of laws/regulations with such requirements are the Fair Credit Reporting Act (FCRA)-USA and the European Union's General Data Protection Regulation (GDPR). In section 609(f) of the FCRA, credit reporting

agencies are required to disclose to consumers the key factors that affected the consumer's credit score. Article 13 (2)(f) of the GDPR requires that in the case of the existence of automated decision making, meaningful information about the logic involved in the processing should be provided to the data subject.

Explainable Ai Approaches

In this section, we will briefly describe some of the approaches used to actualize explainability in AI systems.

Explainability by Design

Explainability by design is used to refer to models which are considered interpretable by default. These models are said to be intrinsically interpretable. Such models are said to be inherently explainable in that the user can easily understand the decision criteria of the model (Baker S. & Xiang S., 2023). The following are some of the models that fall into this category:

1. **Linear regression (LR)**

Linear regression is said to be interpretable because the contribution of each of the attribute to the output value can be determined from its coefficient. Line of best fit plots can also be used visualize the correlation between the independent and the output variable. For multivariate linear regression, each independent variable can be explained using individual variable plots. However the interpretability of the LR models will decrease as the number of the predictor variables increase. In addition, the LR model is unable to adequately cater for non-linear relationship and therefore its performance suffers in such cases.

2. **Generalized Additive Models (GAMs)**

These are models which are similar to linear regression in that they try to fit a line to data (Baker S. & Xiang S., 2023). However they are not constrained to linear features a number of non-linear functions added together. Fitted GAM line for each of the attributes considered can be used to provide visual explanations to the end user.

3. **Logistic Regression**

Logistic regression is mostly used for binary classification and it creates a logistic function that best fits the training data. A visual representation of the probabilities generated by the logistic function can be used to show the separation between the negative and positive classes.

4. Decision Trees

Decision trees are usually depicted using an inverted tree like structure. The nodes of the tree represent the decision points in the tree. By following a series of decision points with the tree, you are able to tell why the tree some decision for a selected data point. They are thus very interpretable by default. However, compared to other methods, their performance is poorer and consequently, their used has been declining.

5. k-nearest neighbors

This algorithm makes a decision by considering the output of k data points that are closest to the new data item. It is used both for classification and regression. Their decision is therefore explained by considering the characteristics of the k neighbours. Their explainability however decreases as the value of k increases.

6. Support Vector Machines

In the low dimension case of one two attributes, SVMs are explainable in that the line or hyper plane separating the outputs can be constructed and visualized. Their interpretability however reduces for more than two attributes.

Post-Hoc Explainability

Post-Hoc explainability refers a situation where an explanation is generated for the decision criteria of the model after model creation. The explanation can be local or global. For local explanation, an explanation is generated for a given data point i.e. why the model generated a particular output for that given data point. For global explanation, you try to get the general decision making criteria for the model. Another classification of classification approaches is between model-agnostic and model-specific approaches. Model-agnostic approaches can be used to generate explanations for different model classes. Model-specific approaches only apply to a given class of models only. Table 1.1 shows a categorization of these methods. You will study them in detail in later modules.

Table 1.1: Explainable AI Techniques

Algorithm	Description	Global/Local explanation	Model-agnostic/Model Specific
Permutation Importance	Ranks Features by their Importance	Global	Model Agnostic
Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE) plots	Plots change in a feature values against the output values.	Global	Model Agnostic
Shapley Additive Explanations (SHAP)	Uses features combinations to calculate feature importance	Global and Local	Model Agnostic
Local Interpretable Model-Agnostic Explanations (LIME)	Train local surrogate models to approximate the black-box model	Local	Model Agnostic
Class Activation Mapping	A heatmap of the areas of the image that contribute most to the classification	Local	Model Specific (Convolution Neural Networks)
Counterfactuals	Focuses on how the input would have changed to generate a different output	Local	Model Agnostic
Anchors	Creation of if-then rules that explain the predictions	Global and Local	Model Agnostic

Elements Of Responsible Ai

Baker S. & Xiang S. (2023) have identified six key components (pillars) of responsible AI:

Fairness

Fairness is the desirable quality of an AI system to be free from bias and discrimination. Such a system should treat different categories of people equitably. AI systems have been found to portray biased based on characteristics such as gender, race and economic status.

Robustness

A robust AI system is one that generates accurate results across all potential use cases. Such a system should also be resistant to malicious attacks. Such a system should therefore perform as expected and be able to recover from failure without causing harm. Robustness is also related to reliability which requires that the system does not generate unreliable or spurious results.

Transparency

Transparency is the most closely related to explainability and it is the requirement that "the decisions made by an AI system are able to be described and reproduced" (Baker S. & Xiang S., 2023). The descriptions should be accessible to developers, users and other interested parties including regulators. Transparency is important for the uncovering of issues in the model that can cause bias or failure. It has been argued that there is need to balance transparency with the need to protect privacy and intellectual property rights (IP). The disclosure of the internal working of a model may lead to the reveal of IP which companies might be keen to protect

Accountability

Accountability is related to taking responsibility for decisions made by AI systems or with the assistance of such systems. Rubel et al. (2019) use the term agency laundering for situations where an agency decisions are made in an automated fashion by a system. The agency then refuses to take responsibility for bad outcomes (if any) of the systems decision and lays the blame on the algorithm/system. Regulatory agencies may therefore require that creators and owners of AI systems are held accountable for the decisions of the AI system in ensuring that such systems meet legal and ethical requirements.

Privacy

The privacy requirement pertains to ensuring that data used for training, validation and testing of AI models is not leaked to unauthorized parties. It also requires that AI systems can be reverse engineered or manipulated to reveal sensitive information about individuals. It also includes regulatory requirements related to handling private data such as consent and notification for use and access of privacy risks while collecting data.

Breach of privacy puts the user at risk for instance, the risk of being a target of malicious agents e.g. online crime agents. There are risks of personal embarrassment and injury to reputation. Systems may also discriminate against a user based on inferred private information e.g. the credit worthiness of the user based on their political inclinations. Inferred information may be beneficial to the user in some domains (for instance in medical practice) but may be detrimental in other areas (for instance in job recruitment).

Safety

Safety is important because of the recognition that AI systems are being deployed in critical safety systems. Example of such systems are autonomous driving AI systems, mental health chat bots and medical diagnosis and treatment systems. The AI systems are expected to act in such a way that they do not harm people. It has been argued that a robust systems can prevent harm by preventing failures. Another requirement in relation to safety is the prevention of harm to the environment. Harm to the environment can for instance happen when systems controlling dangerous systems such as nuclear reactors fail.

Reading Material 1

To learn about the Nexus of Explainable AI and Responsible AI , read the following material

- Baker S., Xiang S. (2023). Explainable AI is Responsible AI How Explainability Creates Trustworthy and Socially Responsible Artificial Intelligence. URL: <https://arxiv.org/abs/2312.01555>. (Section 4)

Use chats, forums, journals, wikis, blogs, question/answer, message my teacher to engage others.

- Responsible AI is not possible without explainable AI. Justify the above statement

Post the answers to the above questions in a discussion Forum